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MASTER THESIS

Do Managerial Characteristics Matter?

Evidence on Managerial Ability in Spanish Manufacturing SMEs

Student:

Emilija Živković

Supervisor:

Gabriela Sicilia Suárez, Universitat Autonoma de Barcelona

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Abstract

This thesis aims to investigate the influence of the managerial ability on the firm's efficiency level and the managerial characteristics that drive it. To do that, we use data of small and medium-sized enterprises (SMEs) in the Spanish manufacturing sector from the World Bank Enterprise Surveys (WBES) and we conduct a three-step analysis following Demerjian et al. (2012). In the first step, the firms' efficiency scores are estimated using the Data Envelopment Analysis (DEA). In the second step, these scores are regressed on a set of firm-specific characteristics out of the control of managers (firms' size, sector, age, foreign ownership, export intensity, and innovation level), in order to obtain a measure of the managerial ability. In the third step, we explore the relationship between the managerial ability and four managerial characteristics including gender, experience, ownership, and the utilization of performance-based bonuses. The results reveal that the mean efficiency of the sample firms was 0.65, implying that there is a significant improvement area for almost all of the firms involved in the study. We also identify a strong association between firms' efficiency and managerial ability across all firms. Regarding the drivers of managerial ability, the use of performance-based bonuses is the most significant and consistent predictor of managerial ability. It was found that the influence of the manager experience follows the inverted U-shaped pattern, while the ownership is positively related to the managerial ability, though to a small extent. Additionally, the analysis found that the impact of manager gender depends on whether the manager is also the owner of the firm. To the best of our knowledge, this is the first study that used the residual-based methodology to study the managerial ability of SMEs manufacturing firms in Spain, contributing new evidence on the role of managerial characteristics in an underexplored segment of the economy.

Keywords: managerial ability; technical efficiency; Data Envelopment Analysis; SMEs; Spain; manufacturing

1. Introduction

Firm performance is a matter of considerable interest in economics, as aggregate productivity growth ultimately depends on how efficiently firms' exploit their available resources. Even firms operating in the same industry and facing similar production technologies and input costs may exhibit significant differences in factor intensities and, consequently, in productivity levels (Syverson, 2011). Understanding the sources of these performance differences is therefore essential for identifying the forces that determine why some economies grow faster than others. A natural framework for analysing such differences is provided by the production frontier theory, which defines the maximum amount of output that can be produced from a given set of inputs without waste (Koopmans, 1951; Debreu, 1951).

The observed distance to the production frontier is postulated to be determined by firms' ability to combine their resources effectively through managerial decision-making. According to the upper-echelons theory (Hambrick & Mason, 1984), differences in the individual characteristics of founders and top-level managers shape organizational decisions, and, ultimately, firm performance. Empirical findings, mostly from Europe, support the view that managerial quality is an important source of heterogeneity in firms' ability to use their resources efficiently. Bloom and Van Reenen's (2007), using a sample of medium-size manufacturing firms in the UK, France, Germany, and the US, finds that more structured management practices are associated with greater productivity, profitability, and survival rates, even after controlling for country and industry effects. More closely related to the Spanish context, Díaz and Sánchez (2008) reach similar conclusions in their study of organizational factors and technical efficiency. Specifically, they find that management's ability to configure capital and labor has a significant effect on the technical efficiency of Spanish manufacturing SMEs. Ruiz-Jiménez and Fuentes-Fuentes (2015), in turn, examine the role of managerial practices in innovation performance of Spanish technology-based SMEs, reporting a positive association. Furthermore, From a different approach, Demerjian, Lev, and McVay (2012) propose a residual-based measure of managerial ability that isolates the component of firms' efficiency attributable to managers after accounting for observable firms' characteristics. They find that this measure is positively associated with subsequent executive compensation and with the quality of financial reporting. Overall, these studies suggest that managerial ability accounts for a substantive portion of the variation in firms' performance and can be empirically identified and analyzed. .

This issue is particularly relevant for small and medium-size enterprises (SMEs), which account for a substantial share of the economic activity and employment in most economies. Their ability to transform available resources into outputs therefore has major implications not only for individual organizations, but also for aggregate productivity and economic growth. This is especially important in the case of manufacturing, where production typically involves the coordination of multiple inputs and where managerial decisions concerning the allocation and combination of labor and capital may have particularly significant consequences for efficiency. In Spain, manufacturing accounts for approximately 10.7% of GDP and 9.9% of employment (World Bank, 2025; Fundación Ivie, 2025) and also plays a major role in Spain's external trade, accounting for 19 percent of the GDP (World Bank, 2025). Understanding the determinants of productivity among the Spanish manufacturing SMEs is therefore of considerable economic relevance.

The aforementioned literature review highlights the role of managerial ability in general and, more specifically, the ability of managers to configure organizational resources to maximize output, as a potentially important determinant of firm efficiency. However, evidence based on residual measures of managerial ability remains inexistent in the context of Spanish manufacturing SMEs. This thesis aims to contribute to filling this gap using a sample of 400 Spanish manufacturing SMEs from the World Bank Enterprise Surveys (WBES). The empirical analysis contains three stages. First, firm-level technical efficiency scores are estimated using the Data Envelopment Analysis (DEA) approach. Second, these efficiency scores are regressed on a set of firm-specific characteristics to isolate the residual component of performance associated with managerial ability. Finally, this residual measure is regressed on four managerial characteristics identified in the theoretical framework in order to test the proposed hypotheses.

The remainder of the thesis is structured as follows. Section 2 reviews the literature on the determinants of firms' efficiency and managerial ability, develops a theoretical framework, and presents the hypotheses to be tested. Section 3 describes the sample used for the analysis, as well as the research design and methodology. In Section 4, the research findings are presented. Section 5 provides a set of robustness checks. Finally, Section 6 concludes the thesis, summarizes its main implications, and identifies opportunities for future research.

2. Literature review, theoretical background and hypothesis development

2.1 Firm Performance and technical efficiency

According to Farrell (1957), technical efficiency refers to a firm's ability to obtain the maximum possible output from a given set of inputs, or equivalently, to use the minimum quantity of inputs required to produce a given level of output. With his seminal work, Farrell's (1957) set the theoretical foundation for measuring productive efficiency relative to a best-practice frontier, rather than relative to an average or predicted level of performance. After Farrell's concept, Charnes et al. (1978) developed Data Envelopment Analysis (DEA). DEA is a non-parametric linear programming technique that constructs an empirical efficiency frontier from the observed input-output combinations of a set of comparable decision-making units (DMUs), without requiring ex-ante assumptions about the functional form of the underlying production technology. Later on, Banker et al. (1984) extended the original constant returns to scale (CRS) formulation to allow for variable returns to scale (VRS). This is particularly relevant when comparing firms of heterogeneous size, as it will be shown in the following study.

One other approach to frontier estimation is Stochastic Frontier Analysis (SFA). This approach has also been proposed by both Aigner et al. (1977) and Meeusen and van Den Broeck (1977). SFA is a parametric technique that divides the deviation from the frontier into two components: a one-sided inefficiency term and a symmetric random noise term capturing measurement error and other stochastic disturbances. As Barasa et al. (2019) note, this distinction allows SFA to separate genuine inefficiency from statistical one.

Both DEA and SFA approaches have their own strengths and weaknesses, and the choice of a particular method depends on the available data, and the research aims as illustrated in comparative studies such as Hjalmarsson et al. (1996). The main advantage of DEA is that it does not require any assumption about the production function, and it can easily deal with multiple inputs and outputs. That is why this approach is especially useful if the true function of production is unknown, which is usually the case when using the sample from the questionnaire with a relatively small size. However, DEA’s major drawback is that it attributes the distance from the best practice frontier to inefficiency, and therefore, it cannot discriminate between inefficiency, imprecision of measurement, or unobserved influences. In addition, DEA efficiency scores are sensitive to extreme values since it estimates the production function from observed data. The key advantage of SFA in comparison to DEA is that it can separate statistical errors from inefficiency and, therefore, it is possible to use hypothesis testing. In addition, SFA is more reliable if there are outliers or extreme values since it is less sensitive to them. However, SFA has a significant limitation that is associated with the necessity to assume a particular functional form of the frontier and distribution of errors at the beginning of the process. This function and distribution cannot be determined by the given data automatically, which is why the researcher should be especially careful to make the right assumptions and choose the suitable method of SFA. By taking into account the advantages and disadvantages of DEA and SFA and considering the characteristics of the study, including the availability of a small sample size and a lack of information about the form of the production function, DEA seems to be a more appropriate approach to estimate firms' technical efficiency.

Since its introduction in the work by Charnes et al. (1978), DEA has become one of the most widespread non-parametric techniques to evaluate efficiency and productivity. According to the bibliographic review by Emrouznejad and Yang (2018), more than 10,000 articles have been published on DEA in scholarly journals by 2016, demonstrating its extensive application area in the analysis of the performance of public and private organizations. More recently, Mergoni et al. (2025) provide a review of DEA's fifty years' development and note its increased application in the analysis of performance in banks, hospitals, farms, education, transport, and local government. Across these applications, DEA is considered a valuable tool in studies that analyze the performance of firms at a particular point in time (Demerjian, et al., 2012), as it does not require the researcher to define the functional relationship between inputs and outputs in advance. This assumption is especially relevant in the current study, as the application of DEA to the survey data may have limited information about the true production function of the population under analysis.

DEA models can also be distinguished by their orientation: an input-oriented model and an output-oriented model. An input-oriented model estimates the level to which inputs could be proportionally reduced without decreasing the level of output (Charnes et al., 1978). An output-oriented model estimates the level to which output could be proportionally expanded given the observed input levels. This study uses the output-oriented model, because, in the short term, SMEs are more likely to try to increase sales with their existing resources than to reduce inputs like labor or capital.

Finally, it is important to note that firm performance is a broad concept that can be captured through several different measures, such as profitability, sales growth, market share, or return on assets, each reflecting a different aspect of how a firm is doing overall. Technical efficiency, as defined by Farrell (1957), is a narrower approach, because it isolates only one specific dimension of this broader concept: how well a firm converts its available inputs into outputs, given its existing technology. It does not capture outcomes that depend on markets or strategic choices, such as pricing power, demand conditions, or the firm's capital structure. Despite this narrower perspective, technical efficiency is still very useful, because it shows how well a firm turns its resources into outputs without being affected by firm size, pricing, or accounting methods, which can make broader profitability measures harder to compare across firms (Demerjian et al., 2012). This is why this study uses technical efficiency scores as the primary firm's performance measure.

2.2 Managerial ability

The efficiency of a firm is determined by multiple factors: partly by forces beyond the control of its managers (like firm size, sector, openness, etc); while the other part is a result of the characteristics and actions taken by their managers. In this research, we refer to this second type of factors as managerial ability..

Managerial ability is commonly defined as a manager's ability to use a firm's given productive resources, such as labor, capital, and materials, more intensively than other managers who operate similar resources (Demerjian et al. 2012). This definition is compatible with the upper echelons concept developed by Hambrick and Mason (1984), which is discussed further in Section 2.3. According to this view, personal characteristics of managers create differences in the decisions they make, which, in turn, lead to different impacts on the firms' performance. The challenge created by the definition of managerial ability is that such ability is unobservable. To resolve this issue, Demerjian et al. (2012) suggest using an indirect approach to operationalize managerial ability. First, they use DEA to estimate firms' efficiency. Second, they regressed this dependent variable on the firms' characteristics that are not directly influenced by managers (i.e., firm size, age, market share). The idea is that efficiency not explained by these factors (residual) is determined by managerial ability.

This logic relies on an assumption worth stating clearly: once a firm's efficiency is purged of size, sector, age, and other structural factors, what remains should mostly reflect managerial decision-making. This assumption has limits. The residual could also pick up measurement error, unobserved firm differences not captured in the second-stage regression, or simply chance. Still, Demerjian et al. (2012) test this assumption in several ways using a rich database. They show the residual is positively linked to later CEO pay, that it predicts more accurate management earnings forecasts, and that it drops after a CEO leaves the firm. These patterns support the idea that the estimated residual captures real managerial ability, not just noise. This approach became an integral part of the studies on managerial ability, which have been conducted subsequently (e.g., Demerjian and Lev, 2021; Anggraini and Sholihin, 2021).

In addition, several articles have established a positive relationship between managerial ability, measured following the Demerjian et al. (2012) residual-based approach, and firm performance. First, Chen, et al. (2021) provide evidence from a large sample of Taiwanese electronics firms that managerial ability significantly predicts firm performance, with capital structure serving as part of the intervening mechanism. Second, according to Ongoro,et al. (2023) in the context of a developing country (Kenya) , managerial ability has also been found to affect firms' investment efficiency; more specifically, firms with higher managerial ability are more likely to employ their financial resources in an efficient manner. Finally, according to one of the most comprehensive assessments of managerial ability, Demerjian and Lev's (2021) overview of the literature, managerial ability, as measured by the residual-based approach, has been positively related to both financial reporting quality and tax efficiency, as well as investment behavior in a variety of settings and industries, and it serves as a determinant of firm performance in general.

Evidence specifically on managerial ability, as measured by the residual-based DEA approach pioneered by Demerjian et al. (2012), in either Spanish or SME manufacturing contexts, appears to be inexistent, as far as we are aware. On the other hand, there is a body of evidence applying different approaches. Díaz and Sánchez (2008) analyse a sample of Spanish SME manufacturing firms with stochastic frontier methods, reporting that organisational variables related to the ability of managers to appropriately adjust capital and labour are associated with technical efficiency levels and that SMEs are less inefficient than their larger counterparts. In a similar vein, but applying different methods, Pérez-Gómez et al. (2018) study the sources of profit efficiency in a sample of Spanish SME manufacturing firms, finding that size, degree of internationalisation, and labour productivity are relevant determinants of it, although they do not identify a variable summarising managerial ability à la Demerjian et al. (2012). Finally, Ruiz-Jiménez and Fuentes-Fuentes (2015) study the role of management capabilities in a set of Spanish technology-based SMEs, finding a positive association with innovative performance, which is exacerbated when top management teams are more gender diverse. Overall, this literature points to the relevance of managerial factors in Spanish SMEs but none of these studies uses the particular approach of measuring managerial ability as a residual in a DEA frontier estimate, which is the novelty of this thesis.

Building on the previous discussion, as well as the literature reviewed above, the following hypothesis is proposed:

H1: Managerial ability, measured by the residuals from the firm-level analysis of technical efficiency, is related to firm performance.

The subsequent subsections disentangle this general hypothesis by specifying and testing particular aspects of managerial ability, including gender, experience, ownership, and managerial practices.

2.3 Managerial characteristics

A large amount of research suggests that firm outcomes are not fully determined by structural or environmental factors alone. Many times, they are also shaped by the characteristics of the individuals who are in top management positions. According to upper echelons theory (Hambrick and Mason, 1984), organizational outcomes, including strategic choices and performance levels, can be partially predicted by the background characteristics of top managers, as these characteristics shape managers' perceptions, interpretations, and consequently their decisions. Building on the general hypothesis outlined above, this section reviews the literature on four managerial characteristics expected to be associated with managerial ability: gender, experience, ownership structure, and managerial practices. Each of these characteristics is discussed individually in the following subsections, together with the specific hypothesis it motivates.

2.3.1 Gender

The correlation between the gender of a firm's manager and its performance often appears inconsistent. Several theories attempt to detail why the gender of a manager can be associated with outcomes in a firm. First of all, risk aversion has been suggested as a possibility: notably, Adams and Funk (2012) analyze the results of a large-scale survey of Swedish company directors, concluding that female directors are generally “less power-oriented and more benevolent than their male counterparts,” with no significant difference in risk aversion, which challenges the popular view of women as overall more risk-averse agents as compared to men. This is one potential mechanism by which women could affect the value and performance of a firm through their managerial role, since power orientation and benevolence, if channeled into managerial action, could shape the distribution of resources within the firm, thus affecting its technical efficiency and, by extension, the metrics used in this thesis to evaluate managerial capability. Yet the evidence on such theory is far from conclusive, as demonstrated by the following two articles.

According to the study conducted by Allison et al. (2023), which used the data of around 130,000 enterprises from 130 countries collected within the World Bank Enterprise Survey, the female-managed firms perform worse on average than the male-managed ones, especially in the case of small and medium-sized enterprises. The scholars explain this finding by the difference in access to credit and the use of digital technologies and more productive workers, which is lower for female-run businesses. However, as Martínez-Zarzoso and Tyrowicz (2023) show in their research, depending on the ownership category, introducing a female leader might have a different effect on the firm performance, making it either higher or lower than the performance of the male-led counterpart. Enterprises with a female-led top management and male-led ownership performed better than the ones with the male ownership and top management. In both Allison et al. (2023) and Martínez-Zarzoso and Tyrowicz (2023) studies, the association between the gender of top-level managers and firm performance was examined, though the performance was not dissected into the productivity and managerial ability, as it is done in this thesis.

A small number of articles have been dedicated to examining the effect of the gender of the manager on the other’s ability to perform managerial activities, specifically using the residual-based approach. Nonetheless, Fernando et al. (2020), who used the American data, report that gender diversity positively contributes to the top management teams’ overall managerial ability, explaining some of the observed associations between gender diversity and firm performance. Baghdadi et al. (2023), in their analysis of women’s presence in the corporate boards, also show that female directors are positively associated with managerial ability, as measured using the approach proposed by Demerjian et al. (2012). Notably, the association becomes evident when women occupy monitoring roles. In turn, an article by Salehi et al. (2026) explores the differences in the managerial ability of female and male CEOs and finds that females tend to demonstrate significantly better performance than their male colleagues. Thus, the three studies, which, unlike the previous three, were primarily focused on firms' financial performance, also provide grounds for suggesting that women's presence and status within the top-level management positions might affect their managerial ability specifically.

Accordingly, based on the findings of Allison et al. (2023), Martínez-Zarzoso and Tyrowicz (2023), as well as the articles mentioned above, the following research hypothesis has been formulated:

H2: The gender of a firm's manager significantly affects its managerial ability.

2.3.2 Experience

The relationship between the experience of managers and the performance of a firm has been extensively analyzed. According to the learning theory, a manager gains domain-specific knowledge through experience, which positively affects their decision-making process- up to certain thresholds (Hambrick & Mason, 1984). The upper echelon theory, also known as the top management theory, posits that a manager's decision-making is affected by their values, which in turn are dictated by their length of experience (Hambrick and Mason, 1984). Furthermore, an inverted U-shaped experience-performance relationship has also been documented. For instance, Huang et al. (2023), through a text-mined analysis of Chinese listed firms' CEOs careers, demonstrate that the depth of a CEO's functional experience has an inverted U-shaped relationship with the performance of the firm, which is explained by the decline in cognitive flexibility associated with too much experience in a similar function. It must be noted that while most scholars agree that there is a positive relationship between experience and performance, it is rarely linear. More specifically, Morales-Solis et al. (2023), by analyzing the data from the World Bank Enterprise Survey of SMEs in emerging markets, discover an inverted U-shaped relationship between a CEO's industry-specific experience and the performance of a firm, with the turning point occurring at approximately eleven years of experience, possibly due to a manager's inability to adapt to changing institutional frameworks, which impairs innovative capacity and thus performance.

It must be noted that none of the mentioned studies directly analyze the link between experience and managerial ability, as this paper attempts to do. The evidence linking experience to a residual-based managerial ability from prior research is limited, although it has been found to have a positive association. Notably, Custódio et al. (2013) take a different angle to analyze the link between experience and managerial ability. The authors construct a general ability measure from CEOs' prior work experience – the number of positions, firms, and industries worked in – and document that they command a premium in terms of compensation, which provides evidence of the value of the experience to her ability. This finding is also supportive of the main hypothesis of this paper. However, since the analysis does not rely on the residual managerial ability (DEA residual) approach, there may be omitted variables. Therefore, this paper hypothesizes that there is a non-linear relationship, likely an inverted U-shape, between the experience of a manager and her managerial ability. As such, the following hypothesis was formulated for this study:

H3: Manager experience has a non-linear ("inverted U") shaped association with managerial ability.

2.3.3 Ownership

The separation of ownership and management is the key point in the definition of agency theory. Jensen and Meckling (1976) explain that agency costs emerge due to the separation of ownership and control when the interests of managers differ from those of owners. The main reason for this difference is that the former is not motivated to maximize the value of the firm (Jensen and Meckling, 1976). Therefore, one might assume that owner-managed firms are more efficient since the principal-agent costs are eliminated due to the unity of decision-making and responsibility. However, there is another view, implying that owner-managed firms are less likely to be efficient because of the lack of professional management, organization, and control. Thus, in owner-managed small and medium-sized enterprises (SMEs), the manager's performance depends on whether the owner-manager is personally involved in the management process. Based on the mentioned theories, the following hypothesis can be formulated:

H4. Whether the top manager is also the firm's largest owner significantly affects the managerial ability.

2.3.4 Managerial practices

Besides individual characteristics of the manager, such as gender, experience, and ownership status, a significant number of works have explored the impact of managerial practices on the firm's performance. By managerial practices, the authors mean the particular way of managing and motivating employees established by managers in the firm and targeting the improvement of the workers' performance. Bloom and Van Reenen (2007) developed and applied a survey instrument to analyze management practices and their impact on productivity, profitability, and sales. The study used the observations from medium-sized manufacturers in the United States, France, Germany, and the United Kingdom. The authors concluded that firms with sophisticated ways of managing workers performed better than those without. As far as the survey is concerned, compensation based on performance is highlighted as a good example of structured management because it provides workers with extra pay for the achieved results and targets the completion of more work than employees normally would do in the absence of such incentives. The study examines the link between performance-based bonuses and productivity and can serve as the basis to establish a connection between structured management practices, in particular the use of performance-based bonuses, and firm efficiency. For this reason, the following hypothesis can be stated:

H5. The use of performance-based bonuses is positively associated with managerial ability.

3. Research design

3.1 WBES Database

The data used for the analysis of this study comes from the World Bank Enterprise Survey (WBES). WBES is a firm survey, covering all sectors of the economy in over 150 countries with more than 207,000 firms surveyed in more than 315 separate country surveys since 2006.[1] The survey is conducted within private enterprises by face-to-face interviews with the owner(s) or top-level administrative management. Information is collected on general characteristics of the enterprise, sales and supplies, management, innovation, finance, labor, and productivity. In this regard, WBES is used by many Spanish firms as a longitudinal survey. In other words, firms are interviewed multiple times over a long period of time and, therefore, the same sample of firms interviewed years ago is re-interviewed, gathering extensive information about the enterprise. This research uses the latest survey from the WBES for Spain applied in 2024, and we focus on the firms of the manufacturing industry.

3.2 Sample data

In this research, we decided to focus on Spanish firms, and more specifically, on small and medium manufacturing firms. As we are aiming to apply DEA for the analysis and efficiency estimates, the selection of firms which will participate in the study is one of the most significant choices to make. As explained in section 2.1, when constructing the efficiency frontier, DEA compares firms to determine which one sets the highest performance frontier for each given input. In doing so, it is essential that the compared firms are relatively homogeneous in terms of their activities, as well as the produced goods and services, and operate with similar production technologies and organizational structure (Dyson et al., 2001). Therefore, in the case of DEA, such a comparison of a manufacturing firm with a retail or financial services firm within the same DEA model would be meaningless, as the inputs would be used for vastly different purposes, resulting in efficiency scores that would be hard to interpret. Therefore, the decision to restrict the sample to firms of the same industry is dictated by the nature of the efficiency analysis.

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[1] Full WBES questionnaire and methodology: https://www.enterprisesurveys.org/en/methodology

As for the choice of the industry, it was dictated primarily by its importance to the national economy. As of 2024, Manufacturing contributes to around 10.7% of Spain's GDP and comprises 9.9% of the workforce (World Bank, 2025; Fundación Ivie, 2025). The sector is of particular strategic importance to the Spanish economy as it accounts for nearly a fifth of the country's Gross Domestic Product (GDP) through exports and the majority of goods exported by Spain are manufactured products (World Bank, 2025). Therefore, analyzing the sources of efficiency and productivity in this sector, as well as examining the influence of managerial attributes identified in section 2.2 and 2.3 on these indicators, is of significant interest. Therefore, the initial sample of companies was formed from 656 firms operating in the Spanish Manufacturing sector.

Before the analysis, the data was screened for missing values. After excluding firms with incomplete information on the input and output variables required to run the efficiency estimations (sales, labor cost, materials cost, capital, and utilities), the sample includes 512 firms.

Moreover the data was screened for potentially invalid observations. Outliers were identified using interquartile range and z-score criteria, as well as a preliminary peer-frequency analysis, which shows how often each firm is used as a benchmark for others in a DEA model. This revealed that 2 firms were outliers and were excluded from the analysis, resulting in a sample of 510 firms[2].

To narrow the analysis to small and medium-sized enterprises (SMEs), the official European Union definition was applied (European Commission, 2003), which classifies an enterprise as an SME if it has fewer than 250 employees and reports annual turnover of no more than €50 million. Both criteria were applied jointly: firm size was measured using the number of permanent, full-time employees reported in the survey, and turnover was measured directly using the reported annual sales variable, already available as the DEA output measure. This approach, combining employment and turnover thresholds to classify firms using WBES-type

[2] One firm was used as a peer for 472 of the remaining 511 firms (92.4 percent of the sample), an unrealistically high proportion. Further testing showed that this firm reported an output-to-input ratio of 91, compared with a ratio of 4.6 for the next best performing firm in the sample. This suggests a data entry error rather than a real efficiency advantage. A second firm was identified similarly, because of its unlikely reported capital value (approximately five billion, against reported annual sales of seven million), which is inconsistent with the firm's other reported characteristics.

data, is consistent with standard practice in studies using Enterprise Survey data (Kuntchev et al.2013). Applying both criteria resulted in a sample of 409 firms.

Since this study also examines the role of managerial characteristics and contextual firm variables in explaining efficiency (Sections 3.3 and 4), firms with missing values on any of the managerial or contextual variables required for the second- and third-stage regressions were also excluded at this stage, rather than after estimating the DEA model. This step removed a further nine firms, leaving a final sample of 400 firms. To keep the DEA estimation and the subsequent regressions fully consistent, the DEA model (Section 3.3) is therefore estimated directly on this final sample of 400 firms[3].

Within this final sample, firms were further classified according to the European Commission's employment-based thresholds into micro (fewer than 10 employees), small (10–49 employees), and medium-sized enterprises (50–249 employees), resulting in 45, 226, and 129 firms in the three categories, respectively.

3.3 Variable selection

The variables used in this study are grouped into three categories: (i) input and output variables used to estimate the efficiency scores at the firm level, (ii) contextual firm-specific characteristics out of the managers' control used in the second stage, and (iii) managerial variables reflecting the characteristics and practices of the firm's top manager. Table 1 summarizes all variables, their definitions, types as well as their corresponding WBES codes.

3.3.1. Input and output variables

Aligned with most of the previous firm-level efficiency analyses (Demerjian et al., 2012; Barasa et al., 2019), the total annual sales were selected as the only output variable to assess the firms' performance. As for the inputs, the study used four variables: labor cost, materials cost, capital cost, and utilities cost. The first two inputs are the largest components of the cost of production in manufacturing firms and represent the costs of manpower and raw materials used in production. Both variables are drawn directly from firms' income statements, as

[3] Excluding these firms before estimating the DEA model was necessary because DEA efficiency scores are relative: each firm's score depends on the specific set of firms it is compared against. Estimating efficiency on a sample of 409 firms and later dropping nine of them for unrelated data reasons would mean that the reported efficiency scores of the remaining 400 firms had been computed against a reference set that no longer matched the final sample used throughout the rest of the analysis.

reported in the WBES questionnaire: labor cost (n2a) corresponds to the question *"please provide the total annual cost of labor including wages, salaries, bonuses, and social security payments*," and materials cost (n2e) corresponds to the question "*please provide the total cost of raw materials and intermediate goods used in production*," both reported for the establishment's last complete fiscal year. Again, following most of the previous research in the field (e.g., Barasa et al., 2019; Demerjian et al., 2012), capital is measured by the cost of all machinery and equipment needed to carry out the productive activities. The data is reported by the following question in the WBES questionnaire (n7a): "*What would be the cost to replace all of this establishment's machinery, vehicles, and equipment with new ones of the same make or equivalent?*" Finally, the utilities cost stands for the electricity, water, and internet expenses and was constructed by combining the related expenditure items reported by the firms (n2b, electricity; n2k, water; n2l, fixed broadband internet).

3.3.2. Firm-specific variables

We have selected six variables, which reflected firms' characteristics unrelated to managerial input but potentially affecting the firms' performance. First, firm size was measured in terms of the number of permanent, full-time employees. Next, there were four categories of firms, which was used to reflect the manufacturing sector in which the organizations operated: *Food, Fabricated Metal Products, Machinery and Equipment, Other Manufacturing*. Third, a dummy variable identified whether an organization had any foreign ownership. Furthermore, the firm age was calculated as the difference between the year of the last completed fiscal year and the year when the firm was founded, establishing organizations' age. Fourth, the export intensity of firms was identified as a percentage of total sales attributed to direct and indirect exports. Finally, innovation is measured as a discrete variable ranging from 0 to 3, reflecting the number of types of innovation activities reported by the firm (new product introduction, new process introduction, and R&D expenditure).

3.3.3 Managerial variables

This research includes four managerial variables to test the hypotheses proposed in Section 2.2. related to the role of managerial characteristics in the managerial ability. First, manager gender was specified as a dummy variable, which was equal to 1 if the firm's top managerial position was held by a female. Next, the manager's experience was measured in years of work in the respondents' sector, with the value of the variable indicating the number of years of work. The measure was selected in compliance with the majority of the previous WBES-related analyses (Morales-Solís et al., 2023). Moreover, a dummy variable identified whether the firm's largest ownership stake belonged to the same individual who held the position of the owner-manager, which was used to operationalize the concept of owner-manager status following Jensen and Meckling (1976). Finally, performance bonus was a dummy variable, which was equal to 1 if an organization reserved a specific bonus for meeting production or sales targets, thus indicating the extent to which the firms relied on the practice of performance-based rewards (Bloom and Van Reenen, 2007).

Table 1. Definitions of DEA, firm-specific, and managerial variables

Variable	**WBES code**	**Definition**	**Type**
DEA input/output variables			
Sales (output)	d2	Total annual sales	Continuous
Labor cost (input)	n2a	Total labor cost	Continuous
Materials cost (input)	n2e	Cost of raw materials and intermediate inputs	Continuous
Capital (input)	n7a	Cost to replace machinery and equipment	Continuous
Utilities (input)	n2b + n2k + n2l	Combined cost of electricity, water, and internet	Continuous
Firm-specific (control) variables			
Firm size	l1	Number of permanent, full-time employees	Continuous
Sector	a4a	Industry sub-sector (Food; Fabricated Metal Products; Machinery and Equipment; Other Manufacturing)	Categorical
Foreign ownership	b2b	Whether the firm has any foreign ownership share	Binary
Firm age	a20y - b5	Difference between last fiscal year and year of establishment	Continuous

Export intensity	d3b + d3c	Percentage of total sales from direct and indirect exports combined	Continuous
Innovation	h1 + h5 + h8	Count (0-3) of innovation activities: new product, new process, R&D expenditure	Count
Managerial variables			
Manager gender	b7a	Whether the top manager is female	Binary
Manager experience	b7	Years of experience of the top manager in the sector	Continuous
Owner-manager	b3a	Whether the largest owner is also the top manager	Binary
Performance bonus	r8	Whether the firm offers bonuses tied to production or sales targets	Binary

Source: Own elaboration based on WBES database.

3.4 Methodology

This study implements a three-stage empirical methodology. In the first stage, the firms' efficiency scores are estimated by using DEA. In the second stage, these scores are regressed on the firm's specific variables, and the residuals from this regression are used as managerial ability estimates. In the third stage, the managerial ability is regressed on the managerial variables of interest. The methodology is similar to the one proposed by Demerjian, Lev, and McVay (2012), who also purge the firm efficiency estimates of the firm's specific factors (out of the manager's control) before relating the changes to the managerial input.

Stage 1: Firm efficiency estimation

Following Charnes et al. (1978) and Banker et al. (1984), an output-oriented, variable returns to scale (VRS) DEA model is estimated. The output orientation implies that the firm is assumed to be interested in increasing its outputs given the available resources. For each firm *k*, the model searches for the maximum proportional expansion of outputs (φ) that is feasible given the firm's observed input levels, relative to the input-output combinations of all other firms in the sample:

$\max \varphi_k$

subject to:

$\Sigma_j \lambda_j x_{ij} \leq x_{ik}$, for each input i=1,…,p. Equation (1)

$\Sigma_j \lambda_j y_{rj} \geq \varphi_k y_{rk}$, for each output r=1,…,q.

$\Sigma_j \lambda_j = 1$

$\lambda_j \geq 0$

where, for the firm *k*, φ_k represents the radial distance between the firm and the frontier, x_{ik} and y_{rk} are inputs and outputs vectors, and λ_j are the virtual intensity weights defining the efficient reference set for firm k. The convexity constraint ($\Sigma_j \lambda_j = 1$) defines VRS, which is more flexible than constant returns to scale and includes varying production scales, which is appropriate given the potentially heterogeneous firm sizes in our sample. The efficiency score for firm *k* is subsequently obtained as $\theta_k = 1/\varphi_k$, which is bounded between 0 (least efficient) and 1 (fully efficient, i.e., located on the efficiency frontier). The greater the score, the greater efficiency. For example, an efficiency score of $\theta_k = 0.8$ means that firm *k* is inefficient and it could improve proportionally all outputs by 25% (1/0.8=1.25).

Stage 2: Managerial ability estimation

The firms' efficiency scores are regressed on firm-specific variables that are not related to managerial decisions to obtain the measure of managerial ability:

$\theta_i = \beta_0 + \beta_1 \text{Size}_i + \beta_2 \text{Size}^2_i + \beta_3 \text{Sector}_i + \beta_4 \text{ForeignOwned}_i + \beta_5 \text{Age}_i + \beta_6 \text{Age}^2_i + \beta_7 \text{ExportIntensity}_i + \beta_8 \text{ExportIntensity}^2_i + \beta_9 \text{Innovation}_i + \varepsilon_i$ Equation (2)

Equation 2 is estimated using Ordinary Least Squares (OLS) following Banker and Natarajan (2008). The quadratic terms of the firm size, age and export intensity are introduced to capture possible non-linearity of their effect on efficiency. The residual from this step, which reflects the part of the firm efficiency unexplained by the observed firm-specific variables, is our measure of managerial ability used as a dependent variable in the third step of the analysis.

Stage 3: Managerial drivers of the managerial ability

The managerial ability (Stage 2 residual) is regressed on the managerial variables corresponding to the hypotheses developed in Section 2.2:

$$Managerial\_ability_i = \gamma_0 + \gamma_1 PerformanceBonus_i + \gamma_2 OwnerManager_i + \gamma_3 Experience_i + \gamma_4 Experience^2_i + \gamma_5 Female_i + \gamma_6 (OwnerManager_i \times Female_i) + u_i \qquad \text{Equation (3)}$$

A quadratic item for the manager's experience is included because of the possibility of the nonlinear connection, as mentioned by Morales-Solís et al. (2023). As for including an interaction item between owner-manager status and manager gender, Martínez-Zarzoso and Tyrowicz (2023) explain that the connection between the manager's gender and the firm's performance depends on whether the owner is managed or not.

Equation (3) was estimated by Ordinary Least Squares (OLS). Multicollinearity was assessed separately for the Stage 2 and Stage 3 models using Variance Inflation Factors (VIF). In the Stage 2 model, VIF values for the linear and quadratic terms of firm size, firm age, and export intensity ranged from approximately 7 to 14. This is expected, since a variable and its squared term are mathematically related by construction, and does not indicate genuine collinearity among substantively distinct predictors; the remaining Stage 2 variables (sector, foreign ownership, and innovation) had VIF values below 1.3. In the Stage 3 model, predictor-based VIF values, which account for the interaction term between owner-manager status and manager gender, were all close to 1.0 (ranging from 1.02 to 1.03), indicating no evidence of multicollinearity among the managerial variables of interest.

4. Empirical results

4.1 The descriptive statistics

This section presents descriptive statistics for the variables used in the three stages of the empirical strategy. Table 2 reports descriptive statistics for the five variables used to construct the DEA efficiency scores: sales (the output variable) and labor cost, materials cost, capital, and utilities (the four input variables).

Table 2. Descriptive statistics of inputs and outputs.

Variable	Mean	SD	Min	Median	Max
Sales	10,429,758	11,731,516	221,000	5,412,640	50,000,000
Labor cost	2,124,406	2,346,395	34,000	1,272,000	12,100,000
Materials cost	5,249,107	7,292,749	5,000	2,085,000	45,000,000
Capital	5,972,890	11,666,063	7,000	2,000,000	120,000,000
Utilities	224,795	1,172,782	0	48,522	22,838,000

Source: own elaboration based on WBES database.

Notes: Sample size n = 400.

Table 2 provides descriptive statistics for the input and output variables used for DEA estimation. The mean of each variable significantly exceeds its median, suggesting that the distribution of each variable is positively skewed. For example, the sales are on average 10.4 million, but the median of the sales is only 5.4 million. Similarly, the capital is on average 6.0 million, but its median is 2.0 million (see Table 1). This implies that despite the screening procedure (Section 3.1), some of the firms in the sample remain large compared to others. For most variables, the standard deviation is considerably large compared to the mean, which shows that there is a large dispersion of data. It is especially evident for the capital and utilities . Other variables also have high variances, which suggests that the firms in the sample have significantly different levels of fixed assets and expenditures on utilities.

Table 3a. Descriptive statistics of firm-specific categorical variables.

Variable	Category	N	%
Foreign ownership	No	360	90.0
	Yes	40	10.0
Sector	Food	62	15.5
	Fabricated Metal Products	106	26.5
	Machinery and Equipment	96	24.0
	Other Manufacturing	136	34.0

Source: own elaboration based on WBES database.

As can be seen from Table 3a, only 10 percent of firms are owned by non-domestic entities. In terms of sector breakdown, the largest category within the sample is *Other Manufacturing*, with 34 percent of firms, followed by *Fabricated Metal Products* (26.5 percent), *Machinery and Equipment* (24.0 percent), and *Food* (15.5 percent).

Table 3b: Descriptive statistics of continuous firm-specific variables.

Variable	Mean	SD	Min	Median	Max
Firm size (employes)	54.39	54.05	4	34	236
Firm age (years)	33.14	19.90	0	31	143
Export intensity (%)	25.79	29.92	0	15	100
Innovation score (0-3)	1.47	1.05	0	1	3

Source: own elaboration based on WBES database.

Table 3b shows that firm age has a mean of 33 years, ranging from newly established firms to one as old as 143 years. The median (31 years) is close to the mean, so the distribution is only slightly skewed.

Export intensity varies widely across the sample, from 0 to 100 percent of sales. The mean is 26% but the median is only 15 %— most firms export at a moderate level, while a smaller group exports intensely.

The innovation score averages 1.47 out of 3, suggesting that most firms engage in one or two innovation activities rather than all three. This is not directly comparable to Barasa et al. (2019), who measure each innovation input separately (26%internal R&D, 41 %formal training, 19% foreign technology adoption) rather than as a combined score, but both studies point to partial rather than comprehensive innovation activity among SME manufacturing firms.

Firm size ranges from 4 to 236 employees, consistent with the EU SME definition used in Section 3.1. The mean (54.39) is well above the median (34), reflecting a right-skewed distribution typical of firm size variables.

Regarding the managerial characteristics (Table 4a), 31.5% of firms offer performance-based bonuses, which means that in the majority of the companies , employees are not paid on a contingent basis. Incentive payments are common throughout most industries, but their incidence is generally lower for small establishments, in line with the findings of Bloom and Van Reenen (2007).

Furthermore, concerning the ownership structure, the majority of firms (61.5%) have an owner as a top-level manager. Thus, in line with the theory of agency costs (Jensen and Meckling, 1976), the separation of ownership and control is rarely practiced in these SMEs since the potential gains from monitoring would be lower than the costs of agency costs, particularly for small firms.

Female top managers are rare in the sample (12%), broadly in line with figures reported for other European economies; for instance, the 2019 WBES for Italy found that 10.5% percent of firms had a female top manager. This suggests that low female representation in top management is not unique to Spain but appears common across comparable European business environments, consistent with the pattern documented by Allison et al. (2023), who find that female-led firms remain a minority across WBES-covered countries generally.

Top managers have an average of 30 years of experience in the sector (SD = 12.40), with the median also at 30 years. This average appears somewhat higher than the figure reported for Italy in the 2019 WBES (mean experience ≈ 26.4 years). This difference likely reflects the fact that the Spanish sample is drawn from a developed economy with an older, more established industrial base, rather than indicating that the sampled managers themselves are old individuals, since this variable captures years of sector-specific experience rather than age.

Table 4a. Descriptive statistics for managerial characteristics categorical variables.

Variable	Category	N	%
Performance bonus	No	274	68.5
	Yes	126	31.5
Owner is manager	No	154	38.5
	Yes	246	61.5
Manager female	No	352	88.0
	Yes	48	12.0

Source: own elaboration based on WBES database.

Table 4b. Descriptive statistics for continuous managerial variables.

Variable	Mean	SD	Min	Median	Max
Managers experience (years)	30.08	12.40	1	30	60

Source: own elaboration based on WBES database.

Table 5 presents the Pearson correlation matrix for the input and output variables, together with a formal test of the monotonicity assumption underlying the DEA model, which requires that each input is positively associated with the output.

Table 5. Pearson's correlation between inputs and outputs.

	Sales	Labor cost	Materials cost	Capital	Utilities
Sales	1.000				
Labor cost	0.811	1.000			
Materials cost	0.895	0.597	1.000		
Capital	0.584	0.446	0.527	1.000	
Utilities	0.209	0.148	0.131	0.149	1.000

Source: own elaboration based on WBES database.
Note: All correlation's coefficients show a p-value < 0.001.

As can be seen from Table 5 all four inputs are positively and significantly correlated with sales ($p < 0.001$ in all cases), which satisfies the monotonicity assumption for DEA estimation: labor cost ($r = 0.811$), materials cost ($r = 0.895$), capital ($r = 0.584$), and utilities ($r = 0.209$). Materials cost is the most correlated with sales, which is expected, as it is the largest component of production costs. The correlation between the input variables is moderate (0.131-0.597), which indicates that, although the variables are related (which is expected, given that larger companies have more of all resources), there is no multicollinearity between them.

4.2 Firms' efficiency and managerial ability

Before proceeding to the analysis of the managerial ability drivers, it is crucial to identify how efficient the firms are. Table 6 and Figure 1 present the descriptive statistics of the DEA efficiency scores and their density distribution, respectively.

According to the DEA model, the mean efficiency score is 0.650 (median 0.626). On average, firms could increase their output by 77.5% using their existing resources to reach the efficient frontier, calculated as the mean of each firm's individual potential improvement (1/efficiency − 1), rather than from the inverse of the mean efficiency score.

The distribution of efficiency scores is concentrated in the upper half of the [0,1] range: 15% of firms (59 out of 400) are fully efficient, an additional 56% achieve efficiency scores between 0.5 and 1.0 without reaching the frontier, and the remaining 29% operate below 0.5, indicating substantial room for improvement.

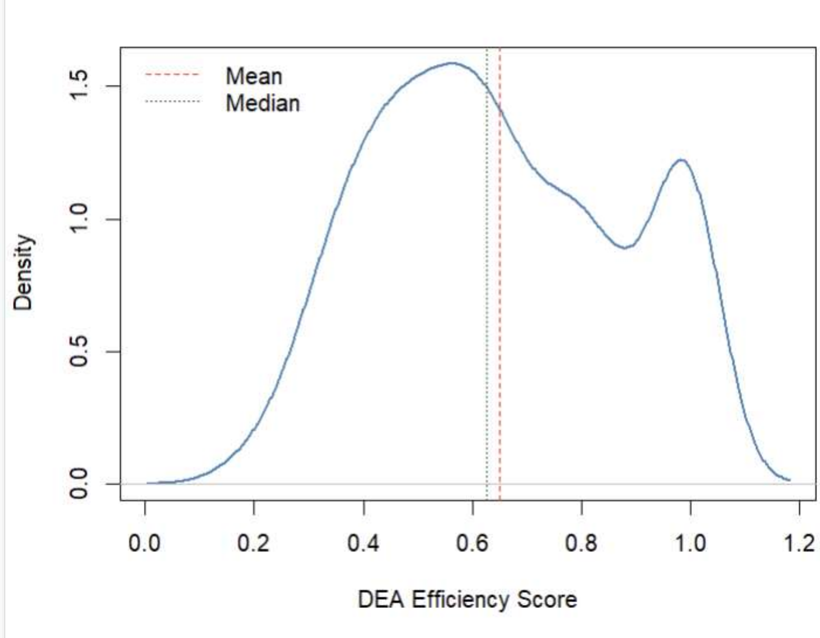
In fact, the distribution of firm's efficiency appears to be right-skewed and bimodal, peaking approximately at 0.5–0.6, representing the central tendency of the observed scores, with the second peak located at the maximum value of 1. This bimodal pattern is a common feature of DEA efficiency scores, which are bounded above by 1 and therefore tend to accumulate firms exactly at the frontier.

Table 6. Descriptive statistics for firms'efficiency scores

Statistic	Value
Mean	0.650
SD	0.225
Min	0.185
Median	0.626
Max	1.000

Source: own estimations based on WBES database.

Figure 1. Firms' efficiency score distribution.



Source: own estimations based on WBES database.

Table 7 presents the results of the second-stage regression analysis. Firms' efficiency scores were regressed on the contextual variables identified in Section 3.3. Overall, the model was significant ($F = 6.772$, $p < 0.001$), but it explained a small fraction of observed variance in efficiency ($R^2 = 0.161$, adjusted $R^2 = 0.137$). It is difficult to say that such a low value is entirely satisfactory, but it is not uncommon either. For instance, Demerjian et al. (2012) report a significantly higher value (37.4%) but also analyze firms' financial performance using a substantially broader set of variables, including financial indicators. Therefore, the present score is lower than one might expect from an analysis based on the more limited data collected from these SMEs.

The size proxy was positively related to efficiency ($p = 0.006$), which is expected and is in line with scale effects of using resources more intensively, consistent with Lundvall and Battese (2000), who similarly find that technical efficiency increases with firm size among manufacturing firms. Furthermore, there was no evidence of non-linearity in the relationship, as the coefficient for the squared size variable was not significant at any conventional level. That implies that the association between size and efficiency holds across all ranges, without any diminishing or increasing returns.

The most intriguing result, perhaps, was that firm age had a negative and significant coefficient ($p = 0.016$). It suggests that, overall, older firms do not seem to perform as well as younger ones in terms of efficiency. However, the squared age variable was positive and significant too ($p = 0.026$). Taken together, these coefficients suggest that the relationship follows a U-shape: firms improve their efficiency as they get older up to a certain point, and then the trend reverses, with older companies becoming less efficient. This pattern is consistent with Coad et al. (2013), who analyze a panel of Spanish manufacturing firms and similarly find mixed age effects on performance, with some firms improving through learning-by-doing while others deteriorate due to organizational inertia. One plausible explanation for this finding is that at some point during the lifespan of a firm, its operations have to be restructured or optimized to adjust to changing circumstances. This adjustment process seems to become increasingly difficult as time goes by, and older firms are less efficient during the years it takes to complete it successfully. In turn, export intensity was positively correlated with efficiency, although the relationship was only marginally significant ($p = 0.068$). The other three variables – sector, foreign ownership, and innovation – were not statistically significant.

Table 7: Second-stage regression: DEA efficiency on contextual variables.

Variable	**Coefficient**	**Std. Error**	**t-value**	**p-value**
Intercept	0.6000	0.0442	13.572	<0.001
Firm size	0.00193	0.00071	2.737	0.006
Firm size²	-0.0000029	0.0000034	-0.851	0.396
Sector: Fabricated Metal	-0.0236	0.0341	-0.692	0.490
Sector: Machinery & Equipment	0.0156	0.0361	0.432	0.666
Sector: Other Manufacturing	0.0150	0.0326	0.461	0.645
Foreign owned	0.0415	0.0379	1.095	0.274
Firm age	-0.00344	0.00142	-2.426	0.016
Firm age²	0.0000322	0.0000144	2.233	0.026
Export intensity	0.00238	0.00130	1.831	0.068
Export intensity²	-0.0000231	0.0000146	-1.587	0.113
Innovation score	-0.00325	0.01059	-0.307	0.759

Source: own estimations based on WBES database.
Note: sample size n = 400.

The residual from the second-stage regression can be thought of as the portion of a given firm's efficiency that cannot be explained by firm-specific characteristics (size, sector, age, etc.). In the next step, the analysis will focus on this residual, which has been interpreted as a proxy for managerial efficiency similarly to the approach used by Demerjian et al. (2012).

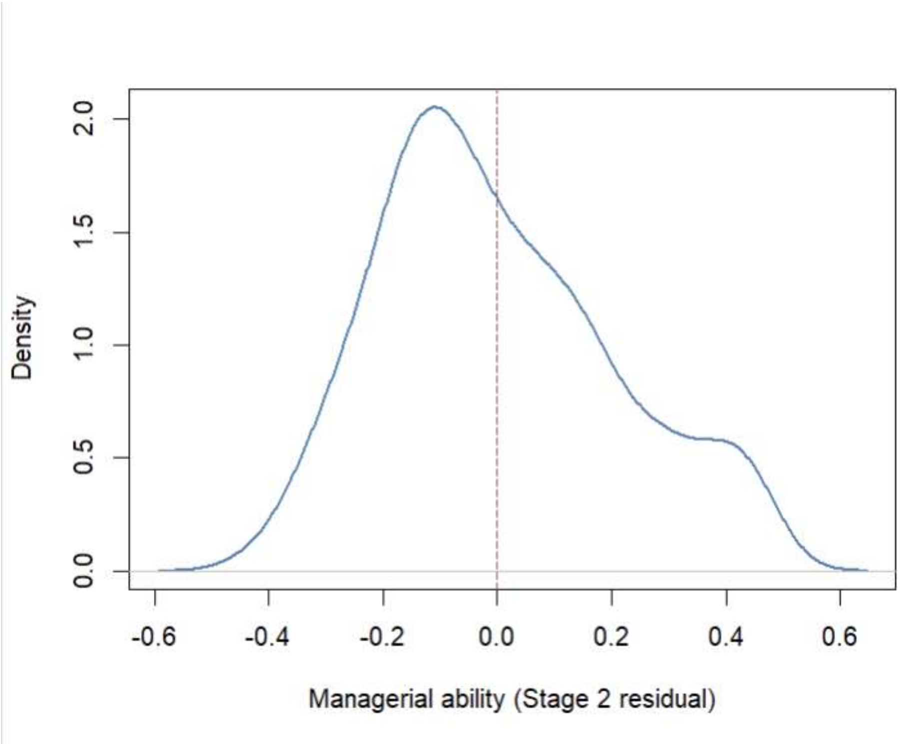
Table 8 and Figure 2 present the descriptive statistics and the density distribution of the managerial ability measure, obtained as the residual from the Stage 2 regression. By construction, the mean of this residual is exactly zero. A positive value therefore indicates that a firm is more efficient than would be expected given its size, sector, age, foreign ownership, export intensity, and innovation activity, while a negative value indicates the opposite. The residual ranges from −0.427 to 0.479, with a standard deviation of 0.206. Slightly more firms fall below the zero benchmark than above it: 226 firms (56.5%) show a negative managerial ability score, while 174 firms (43.5%) show a positive one, consistent with the right-skewed shape of the distribution, where a smaller number of firms achieve unusually high scores while the bulk of the sample clusters modestly below the benchmark. The density plot shows a unimodal, right-skewed distribution, with a single peak located slightly below zero (around −0.05), consistent with the negative median reported above.

Table 8: Descriptive statistics for managerial ability - Stage 2 residual

Statistic	**Value**
Mean	0.000
SD	0.206
Min	-0.427
Median	-0.034
Max	0.479

Source: own estimations based on WBES database.

Figure 2. Distribution of managerial ability across firms.



Source: own estimations based on WBES database.

As hypothesized in H1, firm efficiency and managerial ability are strongly and significantly correlated ($r = 0.916$, $p < 0.001$). This is expected, since managerial ability is, by construction, a component of total firm efficiency. Demerjian et al. (2012) report a similar but weaker correlation ($r = 0.548$). This difference likely comes from the low explanatory power of the Stage 2 model in the present study ($R^2 = 0.161$, compared to 37.4% in Demerjian et al., 2012): since firm context explains less of the variation in efficiency here, the residual retains more of the original efficiency signal, which mechanically produces a stronger correlation.

The strength of this association varies by firm size and age (Table 9). The correlation is strongest among micro firms ($r = 0.996$, N = 45) and weakens for small ($r = 0.974$, N = 226) and medium-sized firms ($r = 0.912$, N = 129). A similar, though weaker, pattern holds for firm age, with a slightly stronger correlation among younger firms ($r = 0.935$, N = 195) than older firms ($r = 0.898$, N = 205). A possible explanation is that larger and older firms are more heterogeneous in the contextual factors captured by the Stage 2 model, so more of their efficiency is explained by observable characteristics rather than by the unexplained managerial component. For micro and younger firms, efficiency is less shaped by these contextual factors and because of that more directly reflected in the residual.

Table 9. Correlation between firm efficiency and managerial ability, by firm size and age.

Group	N	r	p-value
Full sample	400	0.916	<0.001
Micro	45	0.996	<0.001
Small	226	0.974	<0.001
Medium	129	0.912	<0.001
Younger firms	195	0.935	<0.001
Older firms	205	0.898	<0.001

Source: own elaboration based on WBES database.

4.3 The drivers of managerial ability

Table 10 presents the results of the third-stage regression, where managerial ability is regressed on the four managerial variables included in the hypotheses (Section 2.2). The model is significant ($F = 2.686$, $p = 0.014$), but it explains a small amount of variance ($R^2 = 0.039$).

The most notable result is that the coefficient of performance bonus is positive and significant ($p = 0.009$). This finding verifies Hypothesis 5, suggesting that, in settings where a bonus is linked to either production or sales volume, managerial efficiency tends to be higher, net of the influence of other factors. This is consistent with the findings of Bloom and Van Reenen (2007), who report that firms with more formalized management practices are more productive, as well as with earlier evidence from Lazear (2000), who documents substantial productivity gains following a firm's switch from fixed wages to performance-based pay. More recently, Lucifora and Origo (2015) similarly find that performance-related pay is associated with higher firm-level productivity using data from Italian firms, suggesting that this relationship holds across different institutional and national contexts.

The result for the gender of a manager is mixed. While the main effect is not significant ($p = 0.127$), the interaction with the owner-manager dummy is negative and significant ($p = 0.014$). Combining the relevant coefficients shows that female managers who are not owners are associated with a positive, though not statistically significant, effect on managerial efficiency (0.079) relative to male non-owner managers, whereas female owner-managers show a negative combined effect (−0.043), the lowest predicted value among the four possible combinations of gender and ownership. In other words, female owners tend to be less efficient managers, but this pattern does not hold for female managers who are not owners of the firm. This finding partially supports Hypothesis 2 and is consistent with the results of Martínez-Zarzoso and Tyrowicz (2023), who find that a female manager does not necessarily underperform her male colleagues; specifically, they report that firms with a female top manager and exclusively male ownership show higher labor productivity, mirroring the pattern of relatively better performance among female non-owner managers found here.

The effect of owner-manager is consistent with Hypothesis 4, but it is only significant at the 10% level ($p = 0.098$). This result is broadly consistent with agency theory and with Ang et al.(2000), who find that agency costs are lower when a firm is controlled by its owner rather than an outsider, and that asset utilization is positively related to the manager's equity stake, using a sample of small private firms. The weaker significance found here may be because we use a simple dummy variable for owner-manager status as a proxy for equity participation, instead of the actual ownership share used by Ang et al. (2000). This coarser measure likely weakens the strength of the estimated relationship

For the experience of a manager, both the linear and quadratic terms are significant at 10% confidence ($p = 0.064$ and $p = 0.093$, respectively). The positive sign of the linear term and the negative sign of the quadratic term suggest the presence of an inverted U-shaped association, supporting Hypothesis 3. Morales-Solís et al. (2023) find the same pattern. This fits the broader "*too-much-of-a-good-thing*" effect described by Pierce and Aguinis (2013), who show that many positive predictors in management research eventually turn negative past a certain point. One possible explanation is that early experience helps a manager build sector knowledge and make better decisions, which improves efficiency. But after some point, that same experience may lead to overconfidence or a reliance on routines that no longer fit a changing market, and efficiency starts to decline.

To summarize, performance bonus appears to be the most robust managerial practice positively associated with managerial ability. The associations with the other managerial factors are weaker but still present and generally consistent with the predictions of the previous literature and the hypotheses.

Table 10. Third-stage regression: managerial ability on managerial variables.

Variable	Coefficient	Std. Error	t-value	p-value
Intercept	-0.1268	0.0489	-2.594	0.010
Performance bonus	0.0598	0.0226	2.644	0.009
Owner-manager	0.0385	0.0232	1.659	0.098
Manager experience	0.00587	0.00317	1.854	0.064
Manager experience²	-0.0000854	0.0000507	-1.684	0.093
Manager gender (female)	0.0790	0.0517	1.528	0.127
Owner-manager × Manager gender	-0.1609	0.0652	-2.469	0.014

Note: $R^2 = 0.039$; Adjusted $R^2 = 0.025$; $F(6, 393) = 2.686$, $p = 0.014$.

Source: own elaboration based on WBES database.

5. Robustness checks

As a first robustness check, the DEA model was re-estimated without utilities as an input, since this variable showed the weakest correlation with sales among the four inputs (Section 4.1) and is more likely to contain measurement error. The resulting efficiency scores were then used to re-estimate both the second- and third-stage regressions.

The main finding holds up well under this alternative specification. Performance bonus remains positive and significant (0.062, p = 0.010), closely matching the main model (0.061, p = 0.007). The interaction between owner-manager status and gender also remains significant and negative, and is somewhat stronger in this specification (-0.208, p = 0.003, compared to -0.161, p = 0.014 in the main model).

Two other results are more sensitive to this change. Manager experience and its squared term, marginally significant in the main model (p = 0.064 and p = 0.093), lose significance here (p = 0.135 and p = 0.178). Conversely, manager gender, not significant in the main model (p = 0.127), becomes significant in this specification (0.152, p = 0.006). Owner-manager status remains marginally significant in both models (p = 0.098 and p = 0.091).

These results indicate that the central finding of this study, the positive association between performance bonuses and managerial ability, does not depend on whether utilities are included as a DEA input. The results for manager experience and manager gender, however, are more sensitive to this specification choice and should be interpreted with some caution.

As a second test to check if the results were robust, the Stage 2 model was re-estimated using Tobit regression since the dependent variable lies between 0 and 1 and was even censored at the value of 1 for 59 firms. The results were found to be consistent with those obtained using the OLS method. In particular, the coefficient for firm size was positive and significant in both models. The U-shaped influence of the firm age was also found to be significant with the linear component being negative and the quadratic one positive. The results for the export intensity were also positive and significant. No influence was found for the sector, foreign ownership, and innovation in both specifications. Thus, it was found that the results of the Stage 2 regression are not sensitive to the choice of the estimation method and OLS can be used for its simplicity and interpretability.

As a third robustness check, innovation was removed from the Stage 2 model and instead included directly in Stage 3, alongside the managerial variables, since innovation activity could plausibly be considered a managerial rather than a purely contextual characteristic.

The results are highly consistent with the main model. The performance bonus remains positive and significant. The interaction between owner-manager status and gender remains significant and negative at a similar magnitude. Owner-manager status, manager experience, and manager gender all retain coefficients and significance levels very close to those in the main model. Innovation itself is not significant when included in this stage ($p = 0.438$), suggesting that, regardless of whether it is treated as a contextual or a managerial variable, it does not have an independent effect on managerial ability. These results indicate that the main findings are not sensitive to the treatment of innovation as a contextual versus a managerial variable.

6. Conclusions

This thesis addressed a simple question: does managerial ability matter for firm efficiency among Spanish manufacturing SMEs, and if so, which managerial characteristics are associated with it? Using WBES data on 400 Spanish manufacturing SMEs, a three-stage strategy was applied. First, DEA was used to estimate each firm's technical efficiency. Second, these efficiency scores were regressed on firm-specific variables to isolate the part of efficiency associated with managerial ability. Third, this residual measure was regressed on four managerial characteristics: gender, experience, ownership status, and performance-based bonuses.

The results provide mixed but informative support for the proposed hypotheses. Performance-based bonuses emerged as the strongest and most robust factor associated with managerial ability, remaining significant across alternative models considered. This result is consistent with earlier studies linking performance-related pay to productivity (Bloom & Van Reenen, 2007; Lazear, 2000; Lucifora & Origo, 2015) and suggests that this relationship also holds for small manufacturing firms in Spain. Manager experience showed a non-linear, inverted U-shaped relationship with managerial ability, although the effect was only marginally significant, in line with Morales-Solís et al. (2023) in other emerging markets. Owner-manager status was positively, but again only marginally, associated with managerial ability, which is broadly in line with agency theory (Jensen & Meckling, 1976). The results for gender are more complex. No significant direct effect was found, but the interaction between gender and owner-manager status was significant and negative. This suggests that the relationship between gender and managerial ability may depend on whether the manager also owns the firm, rather than being uniform across firms.

The main contribution of this thesis is that, to the best of our knowledge, it is the first study to apply the residual-based measure of managerial ability proposed by Demerjian et al. (2012) to Spanish manufacturing SMEs. This measure has been mostly used with large, listed firms in developed economies, where detailed financial data was available The present thesis shows that the same general approach can be applied to a smaller, survey-based sample of SMEs, extending its use to a segment of the economy that is more difficult to study but represents an important share of the economy. On top of this methodological contribution, the empirical results also provide insights into the role and the determinants of managerial ability.

Performance-based bonuses seem to be the most consistent factor associated with efficiency in this setting, while gender, experience, and ownership status show weaker or more conditional relationships. This finding may be useful for SME owners and policymakers interested in improving productivity in the manufacturing sector.

This study also has several limitations. First, the data are cross-sectional, so the results show associations, not causal relationships. For example, it is possible that more able managers are simply more likely to introduce performance bonuses, rather than bonuses themselves leading to higher managerial ability. Second, both regressions explain only a relatively small share of the efficiency and managerial ability (R^2 of 0.161 and 0.039). This suggests that other unobserved factors, not captured by the WBES questionnaire, may also play an important role. Third, the number of managerial variables available was limited to four: gender, experience, ownership status, and bonus use. Other characteristics that the literature suggests may be relevant, such as managers' education, previous industry experience, risk attitudes, or decision-making style, were not available in the WBES questionnaire and therefore could not be used. Fourth, some of the managerial variables that were available are coarse. Owner-manager status and bonus use, for example, are represented by dummy variables rather than more detailed measures such as ownership share or bonus size, which may limit the ability to capture their relationship with managerial ability. Fifth, the subsample of female-led firms is small (48 out of 400), which limits the strength of the conclusions that can be drawn from the gender results. Finally, the study focuses on a single country and sector, which may limit the generalizability of the findings.

Future research could address these limitations in several ways. Panel data following the same firms across multiple WBES waves, could help to test whether the relationships identified are causal. For example, it would allow researchers to examine whether firms that introduce performance-based bonuses subsequently show improvement in managerial ability. In addition, including other managerial characteristics, such as education or previous experience outside the current sector, could help address some of the omitted-variable concerns raised above. From a methodological perspective, the second- and third-stage regressions used in this research rely on OLS models with linear and quadratic terms, which may not fully capture more complex relationships between firm and managerial characteristics and efficiency. Future research could therefore explore alternative methods, including machine learning techniques such as random forests or gradient boosting, to account for potential non-linearities and interactions between variables. Finally, applying the same approach to other countries and sectors would help determine whether the strong association between performance-based bonuses and managerial ability found in this study is specific to Spanish manufacturing or can be observed more broadly.

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